Aether Engineering Workbench - technical report

Candidate: identified by the ZIP file name. Built 2026-10-31 and 2026-11-01. All numbers below were produced by the code in this repository on the environment named in section 7.

1. Design choices

A desktop workbench (C++17, CMake, Qt 6 Widgets) with a strict split: a Qt-free numerical **core**, a thin Qt-free **application** layer and a **frontend** that only draws and handles input. The interface follows the supplied dark/light palettes; violet marks active navigation, primary actions and selected tools, while green/amber/red appear only for real status. Numbers use a monospaced font. 3-D and 2-D views are software-rendered (QPainter) to avoid GPU-driver dependence. Choices and trade-offs are in `docs/architecture.md`.

2. Architecture

`frontend -> application -> core`. The core holds thermal FE, STL and CSV parsers, batch execution, benchmark, backpropagation, hardware mocks and self-checks. The tests and two command-line tools link only the core and application libraries. The same `solveInto()` routine serves the interactive solver, serial batch and parallel workers, so serial/parallel agreement tests scheduling and memory safety.

3. FEA formulation

Equation -d/dx(kA dT/dx) = qA with L = 1 m, A = 1e-4 m^2, k = 10 W/(m K), q = 1e4 W/m^3, T(0) = 100 C, T(L) = 20 C. Two-node linear elements, Ke = (kA/h)[[1,-1],[-1,1]], fe = (qAh/2) [1,1]. Tridiagonal assembly, Dirichlet conditions by right-hand-side correction plus identity rows, Thomas solve with singularity detection. Invalid parameters raise errors naming the parameter.

4. Analytical verification and mesh convergence

T_exact(x) = 100 - 80x/L + q/(2k) x(L-x). Nodal values are exact to round-off (about 1e-13 C), as expected for linear elements with a uniform source in 1-D. The error is therefore evaluated at element midpoints using FE interpolation.

N	max abs midpoint error [C]	observed order
10	1.25	-
20	0.3125	2.00
40	0.078125	2.00
80	0.01953125	2.00

The values equal q h^2/(8k) exactly. Plots: `results/plots/thermal_profile_*.png`, `thermal_convergence_*.png`. Exports: `results/raw/thermal_*`.

5. Performance results (measured; see `results/benchmarks/benchmark.md`)

Workload: 2000 independent solves, 2000 elements each, fixed seed 20260101, k in [1, 50], q in [1e3, 5e4]. Protocol: one warm-up and five timed repetitions per worker count, median reported, compute only. Machine: virtual Intel Xeon @ 2.10 GHz, **2 logical CPUs**, 7.8 GiB RAM, GCC 13.3.0, Release `-O3`.

Workers	Median [s]	Speedup	Efficiency	max abs diff vs serial
1	0.0718	1.000	1.000	0
2	0.0356	2.018	1.009	0
3, 4	not measured			exceeds the logical CPU count

Agreement criterion 1e-8 C: met (difference 0). An efficiency slightly above 1 on a virtual machine is run-to-run noise or turbo/cache effects and should not be read as super-linear scaling; the 3- and 4-worker rows must be measured on a machine with at least that many logical CPUs. Why speedup can be below p: thread start/join cost, queue contention, workloads too small to keep all workers busy, shared memory bandwidth and caches, hyper-threads sharing a core, other processes. Launch-to-ready is displayed separately in the application.

6. Embedded design

NUCLEO-F446RE (3.3 V I/O, ST-LINK SWD + VCP), Arty A7-35T (3.3 V I/O, FT2232HQ USB-JTAG/UART) and a JTAG-HS3 (Vref 1.8-5 V, drives TCK/TMS/TDI). Research found that the HS3 is not a documented fit for either board directly (Nucleo is SWD-only; Arty J8 is documented for an HS2), which the design states openly. Voltage domains are 3.3 V throughout with a common ground; level shifting is required only for 5 V or 1.8 V peripherals. Items not confirmed from documents are marked UNVERIFIED. Three C++ mock devices (microcontroller, FPGA, JTAG) support discovery, capabilities, connect, disconnect, telemetry and a recoverable timeout; they are labelled SIMULATED in code and UI. Details: `docs/embedded_design.md`.

7. Test outcomes

86 GoogleTest tests passed; 8 of 8 in-application self-checks passed; gradient check maximum error 1.12e-10 (limit 1e-5); loss falls from 1.1825 to 1.08e-20 over 101 updates (1 hand-checkable update plus 100 more). Environment and commands: `docs/testing.md`. Defects found by testing: a segfault caused by a faulty test and a whitespace-only CSV mishandled by the parser; both were fixed. A deliberate challenge test (removing the solver's singular-pivot checks) showed that a backstop guard hid the change from the tests; the test was tightened and now fails in that case (`docs/ai_use_log.md`). The Windows build of the core tool was also run under Wine and reproduced the same convergence table.

Debugging insight. Nodal error is about 1e-13 C and shows no convergence, so verification must use element midpoints, where the error equals q h^2/(8k).

8. Limitations

The Windows GUI package was not built or run here; Launch.exe and the core CLI were only cross-compiled. Benchmark limited to 2 logical CPUs. STL is ASCII-only; CSV nodes are not registered to the STL; the viewport is software-rendered; hardware is simulated; embedded facts partly UNVERIFIED. Full list: `docs/limitations.md`.

9. Future improvements

Build and test on Windows 11 with CI; binary STL; mesh-to-result registration; GPU viewport for large models; ThreadSanitizer runs; real serial adapters (read-only first); non-uniform and 2-D/3-D FE; more benchmark workloads and a persisted benchmark history.

Appendix (2 pages)

Handwritten derivation: print/copy `docs/handwritten_derivation_worksheet.md` by hand, then scan it as the appendix.